



Master in Computer Science
Software Engineering

Cloud Computing

Report 5

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Exercise 1. CloudWatch Dashboard

Question 1. Create a Dashboard for EC2 CPU and Networking Metrics

In this part, we created a CloudWatch dashboard to monitor one EC2 instance.

Steps performed:

1. Logged in to AWS Management Console.
2. Opened CloudWatch and navigated to Dashboards.
3. Clicked **Create dashboard** and provided a dashboard name.
4. Selected widgets and chose **EC2 → Per-Instance Metrics**.
5. Added the required CPU and networking graphs for the target instance.

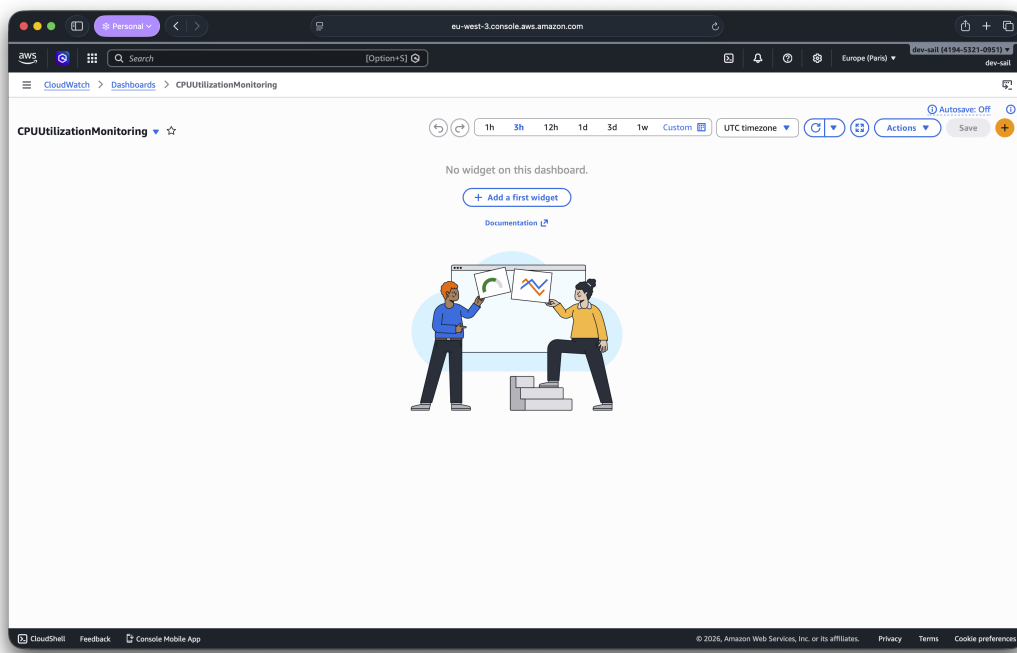


Figure 1: CloudWatch dashboard creation confirmation

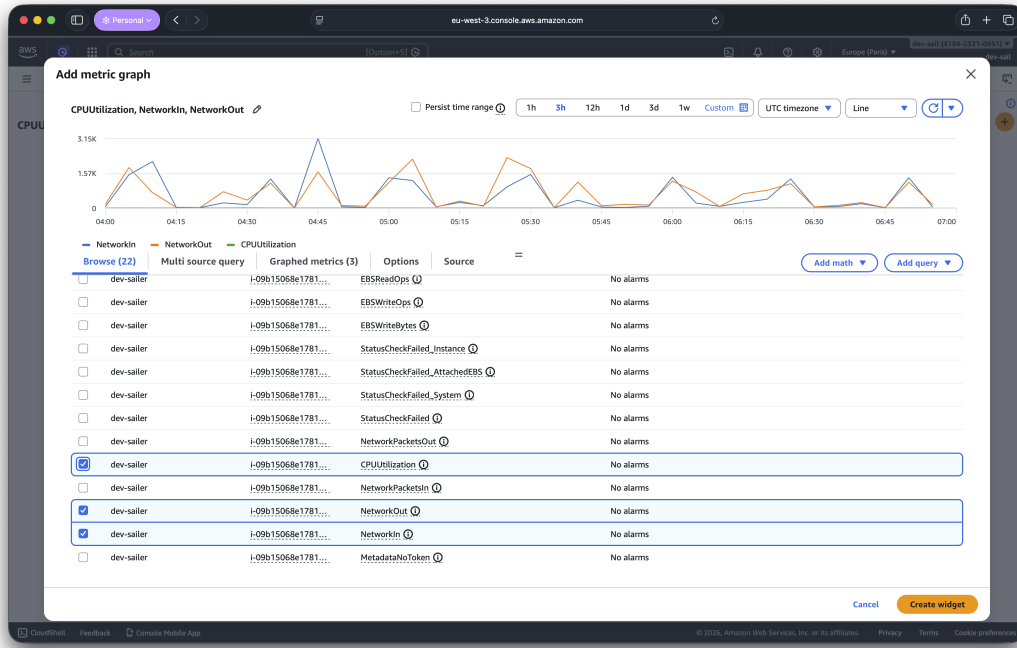


Figure 2: EC2 per-instance metrics selection

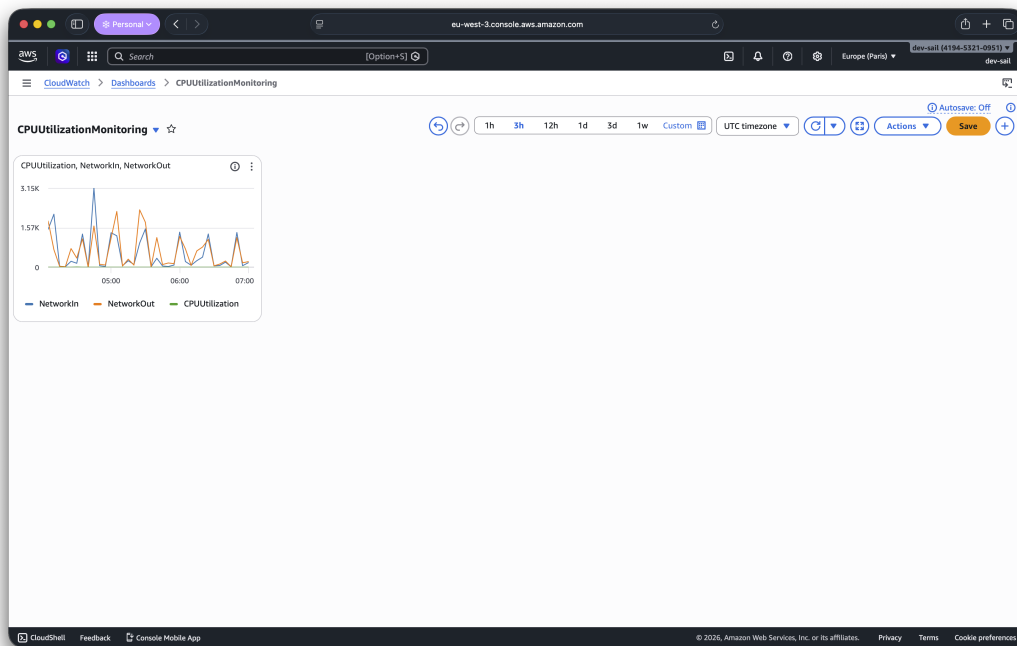


Figure 3: Final dashboard with CPU and network widgets

Question 2. Dashboard Verification

After adding the widgets, the dashboard showed the selected instance metrics correctly.

- CPU utilization is visible on the graph.
- Network In and Network Out are also visible.
- This dashboard gives a quick view of instance health.

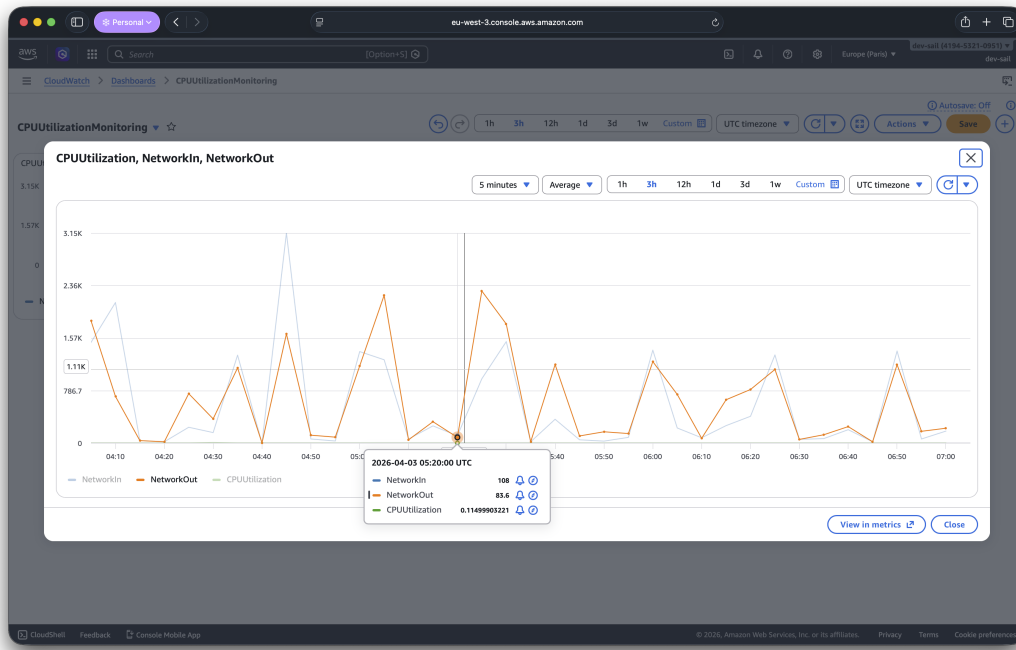


Figure 4: Dashboard runtime monitoring view

Exercise 2. CloudWatch Alarm Creation

Question 1. Create a Billing Alarm for Estimated Charges Above \$500

We created a billing alarm to monitor estimated AWS charges and trigger when the value goes above \$500.

Steps performed:

1. Opened CloudWatch → Alarms.
2. Selected billing metrics and created a new alarm.
3. Set condition type to Static.
4. Defined threshold as Estimated Charges > 500 USD.
5. Reviewed and created the alarm.

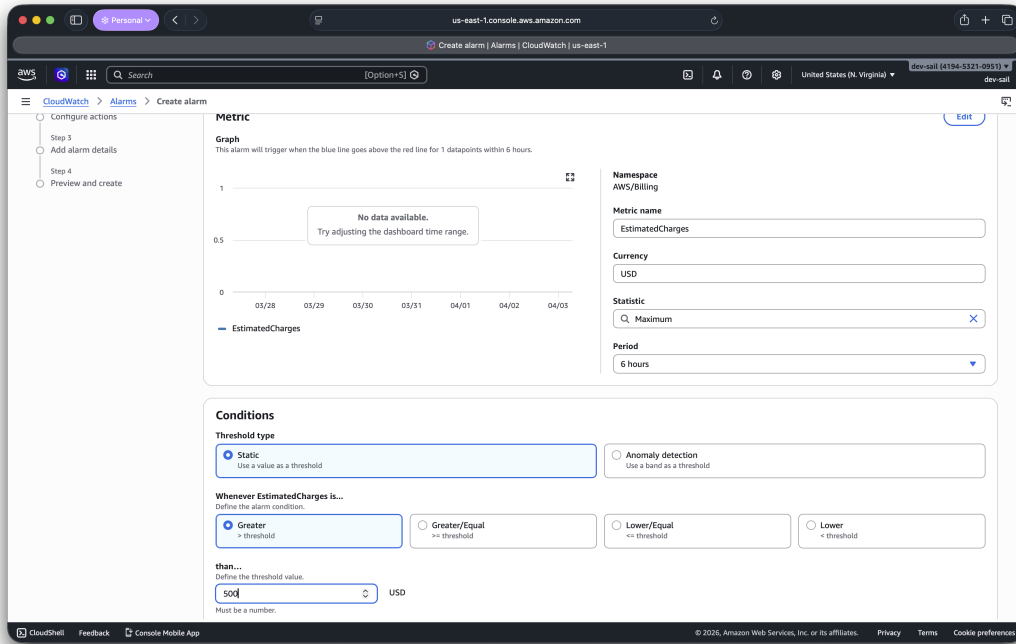


Figure 5: Billing metric selection

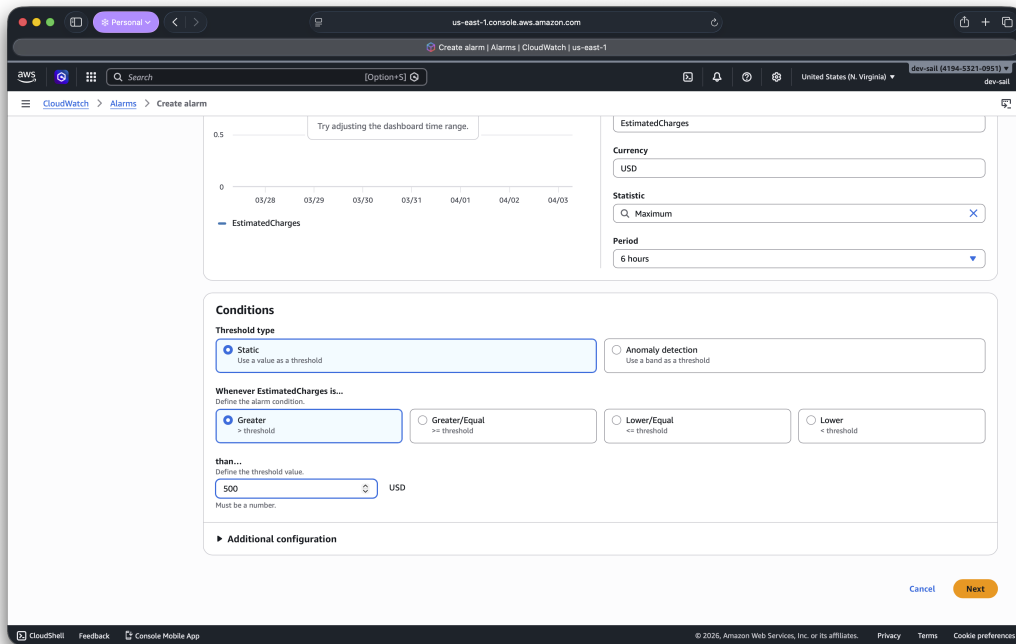


Figure 6: Billing alarm threshold configuration

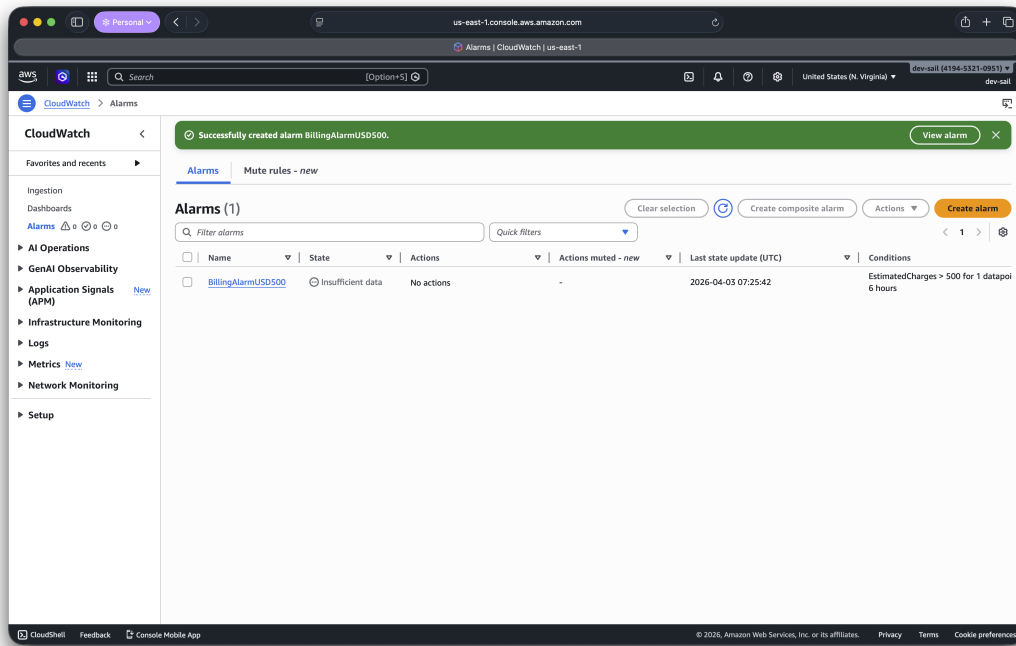


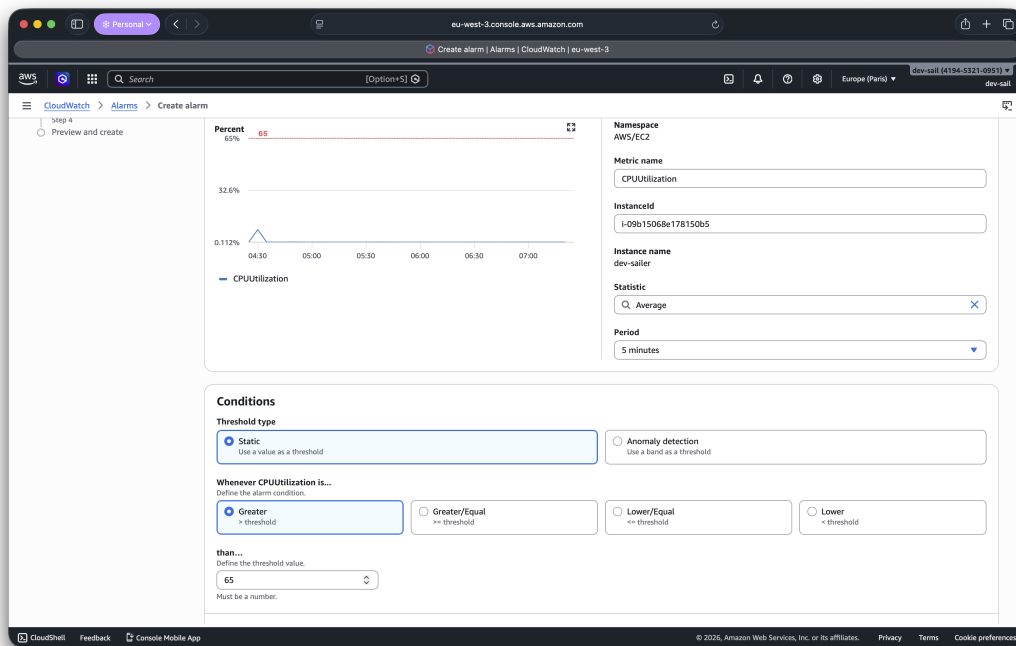
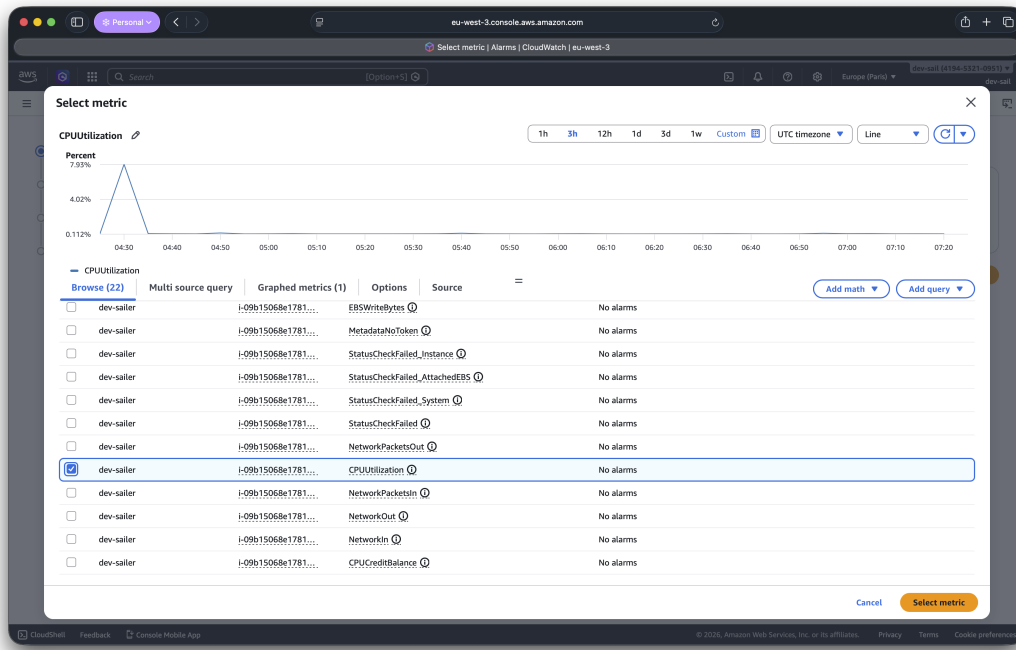
Figure 7: Billing alarm created successfully

Question 2. Create an EC2 CPU Alarm with SNS Notification (65% Threshold)

We also created a CPU utilization alarm for EC2. The alarm changes to **ALARM** when CPU usage is above 65%, and SNS sends an email notification.

Steps performed:

1. Opened CloudWatch → Alarms and clicked Create Alarm.
2. Selected EC2 → Per-Instance Metrics → CPUUtilization.
3. Set threshold to Greater/Equal 65%.
4. Created (or selected) an SNS topic for email notification.
5. Completed alarm name, review, and alarm creation.



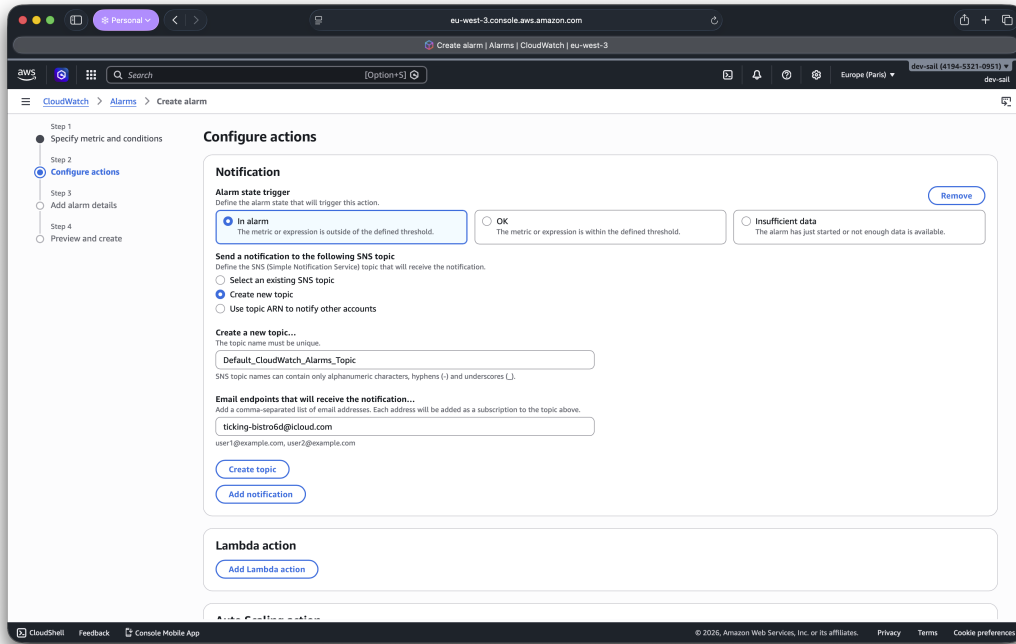


Figure 10: SNS topic configuration for alarm notifications

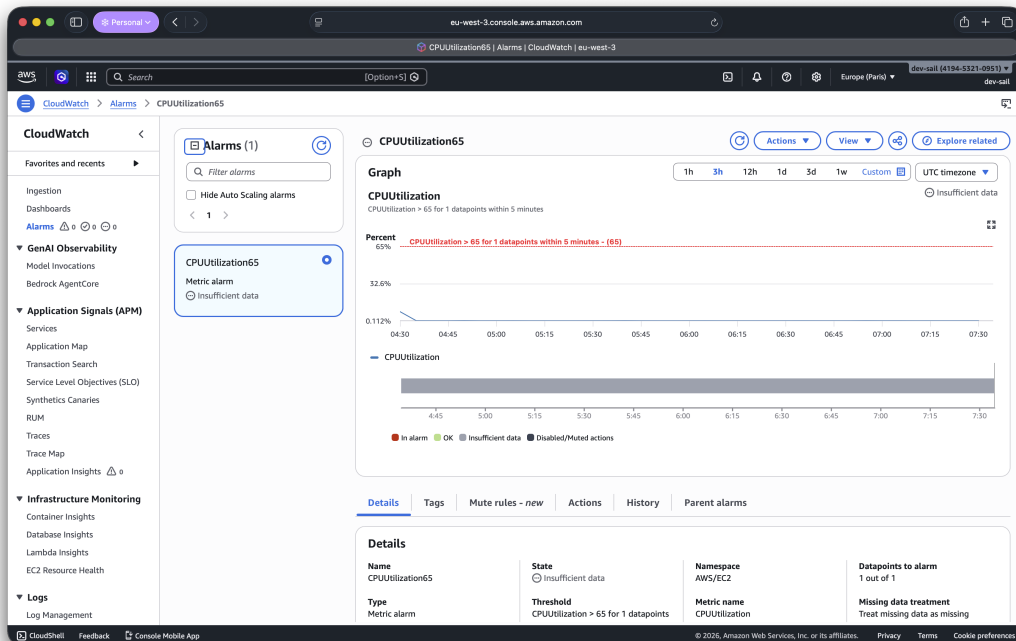


Figure 11: CPU alarm created successfully

Question 3. Email Subscription Confirmation

To receive SNS emails, we confirmed the subscription from the inbox.

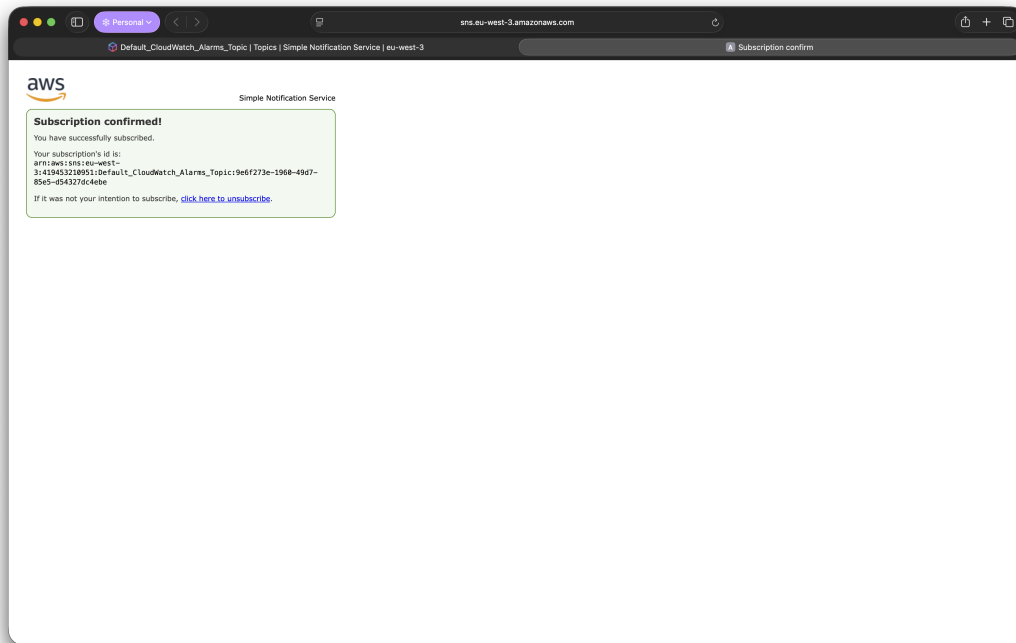


Figure 12: SNS email subscription confirmation

Exercise 3. Conclusion

In this lab, we configured CloudWatch monitoring and alarms. We created a dashboard for EC2 CPU and network metrics, a billing alarm above \$500, and a CPU alarm with SNS email notifications.